

Semiannual Title V Monitoring and Permit Deviation Report

Operating Permit 95OPBO082

AIRS ID 013-0003

CEMEX Lyons Cement Plant

7/1/2023 to 12/31/2023



APPENDIX D

Notification Addresses

1. **Air Pollution Control Division**

Air Pollution Control Division
Operating Permits Unit
APCD-SS-B1
4300 Cherry Creek Drive S.
Denver, CO 80246-1530

ATTN: Blue Parrish – OR – Carissa Money

2. **United States Environmental Protection Agency**

Compliance Notifications:

Office of Enforcement, Compliance and Environmental Justice
Mail Code 8ENF-T
U.S. Environmental Protection Agency, Region VIII
1595 Wynkoop Street
Denver, CO 80202-1129

Note: Deviation reports shall be submitted to the Division at the address given in Appendix D of this permit. No copies need be sent to the U.S. EPA.

SUBMITTAL DEADLINES

First Semi-Annual Monitoring Period: March 1 – June 30

Subsequent Semi-Annual Monitoring Periods: July 1 – December 31, January 1– June 30

Semi-Annual Monitoring Reports: Due August 1, 2008 & February 1, 2009 & subsequent years

First Annual Compliance Period: March 1 - December 31

Monitoring and Permit Deviation Report - Part I

Operating Permit Unit ID	Unit Description	Deviations Noted During Period? ¹		Deviation Code ²	Malfunction/ Emergency Condition Reported During Period?	
		YES	NO		YES	NO
P017	Dowe Flats Quarry Fugitive Emission Sources & Lyons Quarry		X			X
P017	Dowe Flats Quarry – Point Source Emissions		X			X
P018	General Fugitive Emissions Requirements		X			X
P000	Raw Material Storage and Handling at Plant Site	X		9		X
P001	Primary Crusher (Plant Site)		X			X
P002	Raw Materials Drying		X			X
P003	Secondary Crushing, Screen, Silo, Belt Transfer		X			X
P004	Raw Materials Storage Silos		X			X
P005	Raw Mill Grinding		X			X
P006	Homogenizing and Blending		X			X
P007 & P008	Kiln Burning and Clinker Cooling and Transfer to Storage for Finish Mill	X		9		X
P009	Clinker and Gypsum/Additive Silos and Weigh Feeders (Storage and Transfer to Finish Mill)	X		9		X
P010	Sheltered (A-Frame) Clinker Storage and Reclaim	X		9		X
P015	Outdoor Clinker Piles and Handling		X			X
P011 & P012	Cement Finish Mill and Auxiliaries	X		9		X
P013	Cement Silos/Packhouse/Loadout	X		9		X
P014	Material Handling System – Load-In and Load-Out	X		9		X
P007A	Handling and Processing of CKD and Raw Material Waste Dust		X			X
P050	Cement Rail Car Unloading System		X			X
	Gasoline Storage Tank		X			X
	Cold Cleaner Solvent Vats		X			X
LIS-1	Lime storage silo		X			X
LIS-2	Lime weigh hopper		X			X
A-Pit Pump	Diesel fuel-fired engine, rated at 90 hp.		X			X
DF 6" Pump	Diesel fuel-fired engine, rated at 80 hp		X			X
DF 8" Pump	Diesel fuel-fired engine, rated at 84 hp		X			X
Kiln Engine	Natural gas-fired engine, rated at 230 hp (Emergency Engine)		X			X
Flood Engine	Diesel fuel-fired engine, rated at 99 hp (Emergency Engine)		X			X
	Insignificant Activities		X			X
	General Conditions	X		3		X

¹ See previous discussion regarding what is considered to be a deviation. Determination of whether or not a deviation has occurred shall be based on a reasonable inquiry using readily available information.

²Use the following entries as appropriate:

- | | |
|-------------------------|---|
| 1 = Standard: | When the requirement is an emission limit or standard |
| 2 = Process: | When the requirement is a production/process limit |
| 3 = Monitor: | When the requirement is monitoring |
| 4 = Test: | When the requirement is testing |
| 5 = Maintenance: | When required maintenance is not performed |
| 6 = Record: | When the requirement is recordkeeping |
| 7 = Report: | When the requirement is reporting |
| 8 = CAM: | A situation in which an excursion or exceedance as defined in 40 CFR Part 64 (the Compliance Assurance Monitoring (CAM) Rule) has occurred. |
| 9 = Other: | When the deviation is not covered by any of the above categories |

Monitoring and Permit Deviation Report - Part II 8/7/2023 Fresh Air Damper Dusting

Is the deviation being claimed as an: Emergency _____ Malfunction _____ N/A XX

(For NSPS/MACT) Did the deviation occur during: Startup _____ Shutdown _____ Malfunction _____
Normal Operation XX

OPERATING PERMIT UNIT IDENTIFICATION:

AIRs Pt. 031 (P012, S065 – Finish Mill Separator)

Operating Permit Condition Number Citation: 22.15

Explanation of Period of Deviation:

Dusting was present from the Finish Mill Separator's Fresh Air Damper as the fresh air damper remained open after the baghouse was turned off.

Duration (start/stop date & time): 8/7/2023 7:30 – 8:30

Action Taken to Correct the Problem:

After the issue was reported, the finish mill baghouse was turned on which eliminated dusting.

Measures Taken to Prevent a Recurrence of the Problem:

The automated control system's logic was updated to close the fresh air damper prior to shutting down the finish mill baghouse.

Dates of Malfunction/Emergencies Reported (if applicable): N/A

Deviation Code: 9-Other

Division Code QA: _____

Monitoring and Permit Deviation Report - Part II 9/1/2023 – Coal Smoldering

Deviation being claimed as an: Emergency _____ Malfunction _____ N/A XX

(For NSPS/MACT) Did the deviation occur during: Startup _____ Shutdown _____ Malfunction _____
Normal Operation XX

OPERATING PERMIT UNIT IDENTIFICATION: AIRs Pt. 014 (P014 Coal, Clinker and Raw Material Handling System, S020 – Coal Silo/Elevator)

Operating Permit Condition Number Citation: 20.2

Explanation of Period of Deviation:

Compacted coal within the coal silo began to smolder, which resulted in the presence of smoke emerging from the top of the silo.

Duration (start/stop date & time): 9/1/2023 7:35-7:43

Action Taken to Correct the Problem:

Using handheld fire extinguishers along with poking at the bags, the smolder was put out at approximately 7:43am.

Measures Taken to Prevent a Recurrence of the Problem:

Emergency water spouts were installed at the top of the cement silo, which would allow the silo to flood if future smoldering were to occur. Additionally, prior to planned shutdowns of the kiln, procedures are in place to empty the coal silo which mitigates coal compaction.

Dates of Malfunction/Emergencies Reported (if applicable): N/A

Deviation Code: 9 –Other

Division Code QA: _____

Monitoring and Permit Deviation Report - Part II 9/13/2023 – Top of A Frame Dusting

Deviation being claimed as an: Emergency _____ Malfunction _____ N/A XX

(For NSPS/MACT) Did the deviation occur during: Startup _____ Shutdown _____ Malfunction _____
Normal Operation XX

OPERATING PERMIT UNIT IDENTIFICATION: AIRs Pt. 010 (P010 Sheltered Clinker Storage and Reclaim, S051 – Top of A Frame)

Operating Permit Condition Number Citation: Condition 19, 22.15

Explanation of Period of Deviation:

Dusting was observed originating from the grasshopper vent lines connected to dust collector 525-17.

Duration (start/stop date & time): 9/13/2023 07:30-07:42

Action Taken to Correct the Problem:

Operations personnel immediately responded to the scene and identified holes along the vent line, which were repaired to contain dust that was escaping from the system. Additionally, the hopper of dust collector 525-17 was cleaned out to eliminate dust which had the potential to escape the system.

Measures Taken to Prevent a Recurrence of the Problem:

The grasshopper vent lines are monitored on a regular basis to identify and repair any holes that begin to form. Additionally, the hopper of dust collector 525-17 is regularly cleaned.

Dates of Malfunction/Emergencies Reported (if applicable): N/A

Deviation Code: 9-Other

Division Code QA: _____

Monitoring and Permit Deviation Report - Part II 11/14/2023 – Fugitive Dust Event

Is the deviation being claimed as an: Emergency Malfunction ____ N/A ____XX____

(For NSPS/MACT) Did the deviation occur during: Startup ____ Shutdown____ Malfunction ____
Normal Operation ____XX____

OPERATING PERMIT UNIT IDENTIFICATION: AIRS Pt. 007 (P007 Kiln Burning, S016 Precalciner Kiln)

Operating Permit Condition Number Citation: 22.15

Explanation of Period of Deviation:

Dusting was observed from the kiln hood due to a combination of an unstable process, upset chemistry, and the kiln cooling down rapidly.

Duration (start/stop date & time): 11/14/2023 11:40–12:40

Action Taken to Correct the Problem:

Feed was reduced until chemistry and feed issues were corrected. The sweeper truck and the water truck were immediately deployed to clean up material spillage before it could result in further fugitive dusting.

Measures Taken to Prevent a Recurrence of the Problem:

A reminder was given to the control room operators of the need to make quick adjustments when unstable processes arise. Additionally, the electrical team was advised to complete more frequent inspections of the kiln feed system.

Dates of Malfunction/Emergencies Reported (if applicable): N/A

Deviation Code: 9-Other

Division Code QA: _____

Monitoring and Permit Deviation Report - Part II 11/27/2023 – Cement Silo Dusting

Is the deviation being claimed as an: Emergency Malfunction ____ N/A ____ **XX** ____

(For NSPS/MACT) Did the deviation occur during: Startup ____ Shutdown ____ Malfunction ____
Normal Operation ____ **XX** ____

OPERATING PERMIT UNIT IDENTIFICATION: AIRS Pt. 013 (P013 Cement Silos/Packhouses/Loadout)

Operating Permit Condition Number Citation: 22.15

Explanation of Period of Deviation:

Dusting was observed from the top of the southeast end of the cement silos.

Duration (start/stop date & time): 11/27/23 09:10-09:23

Action Taken to Correct the Problem:

Control room was immediately called to shut down the finish mill compressors and waste dust compressors. Dust dissipated approximately 1 minute after the compressors were taken down. To vent the silos, the dust collectors atop the silos were ran to clear any remaining material from the system.

Measures Taken to Prevent a Recurrence of the Problem:

Procedures are in place to complete more frequent checks of the lines atop the cement silos after long shutdowns.

Dates of Malfunction/Emergencies Reported (if applicable): N/A

Deviation Code: 9-Other

Division Code QA: _____

12/7/23 – Fugitive Dust Event

Is the deviation being claimed as an:	Emergency	Malfunction	N/A	XX
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(For NSPS/MACT) Did the deviation occur during: Startup _____ Shutdown _____ Malfunction _____
Normal Operation _____ XX _____

OPERATING PERMIT UNIT IDENTIFICATION: Pt. 024 (P000 Raw Material Storage and Handling at Plant Site)

Operating Permit Condition Number Citation: 22.15

Explanation of Period of Deviation:

With wind gusts reaching above 55 mph, dusting was seen originating from the C-pit and Patio area, as crusts were not properly formed over the material piles. The Kiln and the Raw Material Dryer were not in operation on this day.

Duration (start/stop date & time): 12/7/2023 11:37-12:07

Action Taken to Correct the Problem:

Two water trucks were immediately deployed to create a crust over the raw material stock piles, which eliminated dusting. The trucks ran throughout the day to stay on top of crusting the piles.

Measures Taken to Prevent a Recurrence of the Problem:

Training was held with all operations personnel who are tasked with operating equipment in the C-Pit and Patio areas. This training reinforced actions that need to be taken by all personnel to ensure raw material piles are properly crusted.

Dates of Malfunction/Emergencies Reported (if applicable): N/A

Deviation Code: 9-Other

Division Code QA:

Monitoring and Permit Deviation Report - Part II

12/31/23 -- CEMS Downtime in Excess of 5%

Is the deviation being claimed as an: Emergency Malfunction ____ N/A ____ **XX** ____

(For NSPS/MACT) Did the deviation occur during: Startup ____ Shutdown ____ Malfunction ____
Normal Operation ____ **XX** ____

OPERATING PERMIT UNIT IDENTIFICATION: Monitoring Requirements

Operating Permit Condition Number Citation: 22.30

Explanation of Period of Deviation:

Quarter 4 CEMS downtime exceeded 5% for CO, NO_x, SO₂, and Flow parameters. These downtimes were due to a stack flow daily calibration failure.

Duration (start/stop date & time): Quarter 4, 2023

Action Taken to Correct the Problem:

Manually calibrated flow and cleaned transducers.

Measures Taken to Prevent a Recurrence of the Problem:

A CEMS replacement is on order.

Dates of Malfunction/Emergencies Reported (if applicable): N/A

Deviation Code: 3-Monitor

Division Code QA: _____

Monitoring and Permit Deviation Report - Part II

11/16/23 – 2 APENs Submitted Late

Is the deviation being claimed as an: Emergency Malfunction ____ N/A ____ **XX** ____

(For NSPS/MACT) Did the deviation occur during: Startup ____ Shutdown ____ Malfunction ____
Normal Operation ____ **XX** ____

OPERATING PERMIT UNIT IDENTIFICATION: General Permit Conditions

Operating Permit Condition Number Citation: Section IV 22.e

Explanation of Period of Deviation:

APEN 014 and APEN 050 were submitted late.

Duration (start/stop date & time): Quarter 4, 2023

Action Taken to Correct the Problem:

APENs were submitted on 11/16/2023.

Measures Taken to Prevent a Recurrence of the Problem:

The site has employed better tracking regarding due dates.

Dates of Malfunction/Emergencies Reported (if applicable): N/A

Deviation Code: 6-Record

Division Code QA: _____

Monitoring and Permit Deviation Report - Part III

REPORT CERTIFICATION

SOURCE NAME: CEMEX Construction Materials South, LLC – Lyons Cement Plant

FACILITY IDENTIFICATION NUMBER: 013-0003

PERMIT NUMBER: 95OPBO082

All information for the Title V Semi-Annual Deviation Reports must be certified by a responsible official as defined in Colorado Regulation No. 3, Part A, Section I.B.38. This signed certification document must be packaged with the documents being submitted.

STATEMENT OF COMPLETENESS

I have reviewed the information being submitted in its entirety and, based on information and belief formed after reasonable inquiry, I certify that the statements and information contained in this submittal are true, accurate and complete.

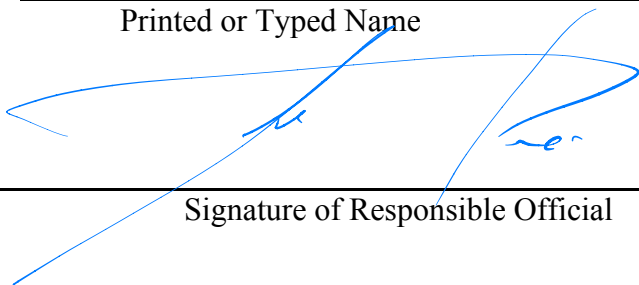
Please note that the Colorado Statutes state that any person who knowingly, as defined in Sub-Section 18-1-501(6), C.R.S., makes any false material statement, representation, or certification in this document is guilty of a misdemeanor and may be punished in accordance with the provisions of Sub-Section 25-7 122.1, C.R.S.

Mr. Erasmo "Erik" Estrada Tovar

Plant Manager

Printed or Typed Name

Title



Signature of Responsible Official

1/31/24

Date Signed

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